



Natural Language Processing

Lecture

**Evaluating Hypothesis
(Models)**

Author:

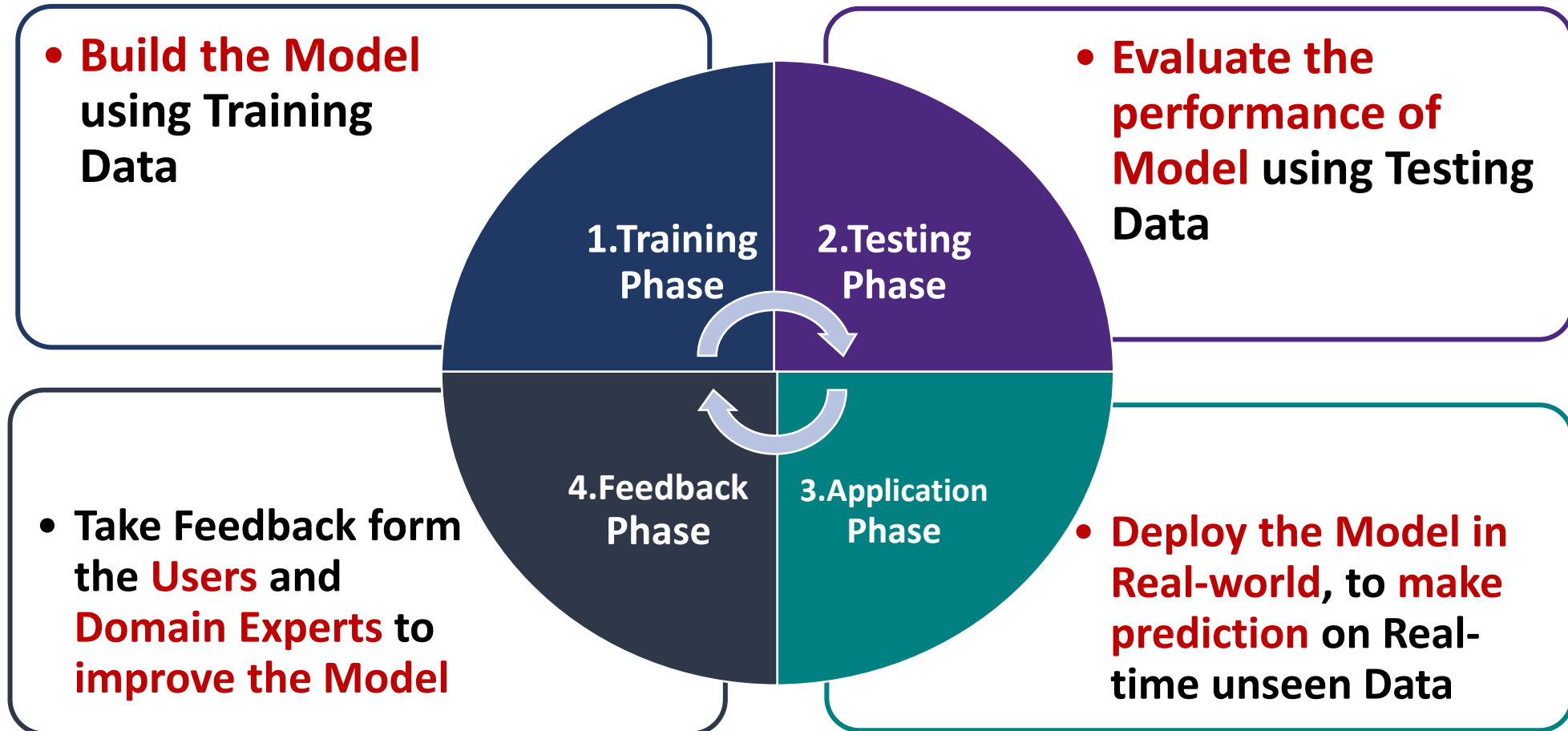
Dr. Rao Muhammad Adeel Nawab

Instructor:

Mr. Waqar Ashiq

Recall The Machine Learning Cycle

Four phases of a Machine Learning Cycle are



Lecture Focus

A **Model is evaluated** in **Testing Phase** using Testing Data

Evaluation is **quantified** using
☐ Evaluation Measures(s)

Therefore, this Lecture will **only focus** on
☐ Testing Phase of Machine Learning Cycle

Why Evaluate Hypotheses (Model)?



Why Evaluate Hypothesis (Model)? (Cont.)

Question

- ❑ Why should we **evaluate h (Model)**?

Answer

- The **main goal** of **building a Model** (Training / Learning) is to **use** it in doing **Real-world Tasks** with **good Accuracy**
- No one can **perfectly predict**, that if a Model performs well in **Training Phase**, then it will also perform well in **Real-world**
- However, **before deploying a Model** in the Real-world, it is **important** to know
 - How **well** will it **perform** on **unseen** Real-time Data?

Evaluation Measures for Classification Problems



Evaluation Measures for Classification Problem

□ Some of the **most popular** and **widely used** Evaluation Measures for Classification Problems are

- Accuracy
- Precision
- Recall
- F_1
- Area Under the Curve (AUC)

Accuracy

Definition

- Accuracy is defined as the proportion of correctly classified Test instances

Formula

$$\text{Accuracy} = \frac{\text{Number of Correctly Classified Test Instances}}{\text{Total Number of Test Instances}}$$

Note

$$\text{Accuracy} = 1 - \text{Error}$$

Accuracy (Cont.)

Question

- When it will **more suitable** to use Accuracy measure?

Answer

- Accuracy evaluation measure is **more suitable** to use for **evaluation** of Machine Learning Algorithms when we have
 - ❑ **Balanced Data**

Example – Calculating Accuracy



Problem Description

Rasheed wants to evaluate his proposed Machine Learning Algorithms (called Excellent Learner) for the task of counter terrorism Identification. His Test Data comprises of 600 instances (300 Terrorism and 300 Non-Terrorism). Rasheed applied Excellent Learner on Test Data and 500 instances were correctly classified. What is the Accuracy?

Example – Calculating Accuracy (Cont.)

Calculating Accuracy for Efficient Learner (Machine Learning Algorithm)

$$\text{Accuracy} = \frac{\text{Number of Correctly Classified Test Instances}}{\text{Total Number of Test Instances}}$$

$$\text{Accuracy} = \frac{500}{600}$$

$$\text{Accuracy} = 0.83$$

Strengths and Weaknesses - Accuracy

□ We can **evaluate** and **compare** various Machine Learning Algorithms using Accuracy evaluation measure

Accuracy fails to **accurately** evaluate a Machine Learning Algorithm when Test Data is **highly unbalanced**

Example – Accuracy is a Poor Measure for Highly Unbalanced Data

- ❑ Consider a Binary Classification Problem with two classes: Positive and Negative. Test Data comprises of 1000 instances, out of which 995 instances are Negative and 5 are Positive.
- ❑ A Machine Learning Algorithm which **always predicts Negative**, will have an
 - Accuracy of 0.995

Example – Accuracy is a Poor Measure for Highly Unbalanced Data (Cont.)

❑ Problem

- Machine Learning Algorithm has **very high** Accuracy (99.5%) on Test Data, even though it **never correctly predicts Positive Test Examples**

❑ A Possible Solution

- Confusion Matrix

Confusion Matrix

Definition

A Confusion Matrix is a table used to describe the **performance of a Classification Model** (or Classifier) on a Set of Test Examples (Test Data), whose **Actual Values** (or True Values) are **known**

Confusion Matrix

Purpose

- ❑ To get **deeper insights** into Model / Classifier **behavior**

Advantages

- ❑ Confusion Matrix **allows** us to **visualize** the **performance** of a Model / Classifier
- ❑ Confusion Matrix **allows** to **separately** get **insights** into the **Errors** made by **each Class**
- ❑ Confusion Matrix gives insights to **both**

Confusion Matrix (Cont.)

Advantages

- **Errors** made by a Model / Classifier and
- **Types of Errors** made by a Model / Classifier
- ❑ Confusion Matrix **allows** us to **compute** many **different Evaluation Measures** including
 - Accuracy
 - Precision
 - True Positive Rate (or Recall)
 - True Negative Rate
 - False Negative Rate
 - False Positive Rate
 - F_1

Confusion Matrix

Confusion Matrix for a Machine Learning Problem with **n** Number of Classes is given below,

		Predicted				
		Class 01	Class 02		Class n-1	Class n
Actual	Class 01					
	Class 02					
						
	Class n-1					
	Class n					

Confusion Matrix for Concept Learning

Confusion Matrix for Concept Learning (a.k.a. Binary Classification Problem) is given below,

		Predicted	
		Class 01	Class 02
Actual	Class 01	a	b
	Class 02	c	d

Confusion Matrix for Concept Learning (Cont.)

□ Considering that

- Class 01 = Non-Terrorism
- Class 02 = Terrorism

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	a (TP)	b (FN)
	Terrorism	c (FP)	d (TN)

Extracting Various Evaluation Measures from Confusion Matrix

□ Next Slides I will try to explain how to extract the following Evaluation Measures from Confusion Matrix

- Accuracy
- Precision
- True Positive Rate (or Recall)
- True Negative Rate
- False Negative Rate
- False Positive Rate
- F_1

Extracting Various Evaluation Measures from Confusion Matrix (Cont.)

Accuracy

Accuracy (AC) is the proportion of the total number of predictions that were correct

$$AC = \frac{(TP) + (TN)}{(TP) + (TN) + (FP) + (FN)}$$

□ This is the same as Accuracy evaluation measure defined earlier

Evaluation Measures from Confusion Matrix (Cont.)

Recall or True Positive Rate (TPR) or Sensitivity

Recall or True Positive Rate (TPR) or Sensitivity is the proportion of Positive cases that were correctly classified

$$TPR = \frac{(TP)}{(FN) + (TP)}$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	a (TP)	b (FN)
	Terrorism	c (FP)	d (TN)

Evaluation Measures from Confusion Matrix (Cont.)

True Negative Rate or Specificity

True Negative Rate (TNR) or Specificity is defined as the proportion of Negatives cases that were classified correctly

$$TNR = \frac{a(TN)}{a(TN) + b(FP)}$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	a (TP)	b (FN)
	Terrorism	c (FP)	d (TN)

Evaluation Measures from Confusion Matrix (Cont.)

False Positive Rate

False Positive Rate (FPR) is the proportion of Negative cases that were incorrectly classified as Positive

$$FPR = \frac{(FP)}{(FP) + (TN)}$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	a (TP)	b (FN)
	Terrorism	c (FP)	d (TN)

Extracting Various Evaluation Measures from Confusion Matrix (Cont.)

False Negative Rate

False Negative Rate (FNR) is the proportion of Positive cases that were incorrectly classified as Negative

$$FNR = \frac{(FN)}{(FN) + (TP)}$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	a (TP)	b (FN)
	Terrorism	c (FP)	d (TN)

Extracting Various Evaluation Measures from Confusion Matrix (Cont.)

Precision

Precision (P) is the proportion of the predicted Positive cases that were correct

$$P = \frac{d(TP)}{b(FP) + d(TP)}$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	a (TP)	b (FN)
	Terrorism	c (FP)	d (TN)

Trade Off Between Precision and Recall

Problem

- There is a **trade-off** between Precision and Recall

A Possible Solution

- Find out Evaluation Measures which **efficiently combine** Precision and Recall
- Example
 - F-measure (combines Precision and Recall)

F₁-measure

F₁-measure

When we assign **same weights** to Precision and Recall i.e. $\beta = 1$, the F-measure becomes F₁-measure

$$F_1 = \frac{2 * Precision * Recall}{Precision + Recall}$$

Example 01 – Confusion Matrix for Binary Classification Problem (Cont.)



Confusion Matrix

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	300	100
	Terrorism	300	300

Example 01 – Confusion Matrix for Binary Classification Problem (Cont.)



Accuracy

$$AC = \frac{a + d}{a + b + c + d}$$

$$AC = \frac{300 + 300}{300 + 100 + 300 + 300}$$

$$AC = \frac{600}{1000}$$

$$AC = 0.6$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	TP 300 a	FN 100 b
	Terrorism	FP 300 c	TN 300 d

Example 01 – Confusion Matrix for Binary Classification Problem (Cont.)



True Positive Rate (TPR) or Recall or Sensitivity

$$TPR = \frac{TP}{TP + FN}$$

$$TPR = \frac{300}{300 + 100}$$

$$TPR = \frac{300}{400}$$

$$TPR = 0.75$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	TP 300	FN 100
	Terrorism	FP 300	TN 300

Example 01 – Confusion Matrix for Binary Classification Problem (Cont.)



True Negative Rate (TNR) or Specificity

$$TNR = \frac{TN}{TN + FP}$$

$$TNR = \frac{300}{300 + 300}$$

$$TNR = \frac{300}{600}$$

$$TNR = 0.50$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	TP 300	FN 100
	Terrorism	FP 300	TN 300

Example 01 – Confusion Matrix for Binary Classification Problem (Cont.)



False Positive Rate (FPR)

$$FPR = \frac{FP}{FP + TN}$$

$$FPR = \frac{300}{300 + 300}$$

$$FPR = \frac{300}{600}$$

$$FPR = 0.50$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	TP 300	FN 100
	Terrorism	FP 300	TN 300

Example 01 – Confusion Matrix for Binary Classification Problem (Cont.)



False Negative Rate (FNR)

$$FNR = \frac{FN}{FN + TP}$$

$$FNR = \frac{100}{100 + 300}$$

$$FNR = \frac{100}{400}$$

$$FNR = 0.25$$

		Predicted	
		Non-Terrorism	Terrorism
Actual	Non-Terrorism	TP 300	FN 100
	Terrorism	FP 300	TN 300

Example 01 – Confusion Matrix for Binary Classification Problem (Cont.)



Precision

$$P = \frac{TP}{TP + FP}$$

$$P = \frac{300}{300 + 300}$$

$$P = \frac{300}{600}$$

$$P = 0.50$$



F₁ Measure

$$F_1 = \frac{2 * Precision * Recall}{Precision + Recall}$$

$$F_1 = \frac{2 * 0.5 * 0.75}{0.5 + 0.75}$$

$$F_1 = \frac{0.75}{1.25} = 0.6$$

Example 02 – Confusion Matrix for Ternary Classification Problem



□ Consider the following Machine Learning Problem

Irfan had a collection of 2000 documents, which can be categorized as Positive (1000 documents), Negative (600 documents) and Neutral (400 documents). He applied a Naïve Bayes classifier on it. Out of 1000 Positive documents, half were correctly classified and 300 were classified as Neutral. For Negative class, 150 were classified as Positive and 250 as Negative. In case of Neutral class, 300 were correctly classified and 50 were classified as Negative.

Example 01 – Confusion Matrix for Ternary Classification Problem (Cont.)



Confusion Matrix

		Predicted		
		Positive	Negative	Neutral
Actual	Positive	a	b	c
	Negative	d	e	f
	Neutral	g	h	i

Example 01 – Confusion Matrix for Ternary Classification Problem (Cont.)



Confusion Matrix

❑ Total Documents = 2000

❑ Positive Documents = 1000

❑ Negative Documents = 600

❑ Neutral Documents = 400

		Predicted		
		Positive	Negative	Neutral
Actual	Positive	500	200	300
	Negative	150	250	200
	Neutral	50	50	300

Example 01 – Confusion Matrix for Ternary Classification Problem (Cont.)



Accuracy

$$AC = \frac{a + e + i}{a + b + c + d + e + f + g + h + i}$$

$$AC = \frac{500 + 250 + 300}{500 + 200 + 300 + 150 + 250 + 200 + 50 + 50 + 300}$$

$$AC = \frac{1050}{2000}$$

$$AC = 0.525$$

Example 01 – Confusion Matrix for Ternary Classification Problem (Cont.)



Confusion Matrix

❑ Total Documents = 2000

❑ Positive Documents = 1000

❑ Negative Documents = 600

❑ Neutral Documents = 400

		Predicted		
		Positive	Negative	Neutral
Actual	Positive	500	200	300
	Negative	150	250	200
	Neutral	50	50	300

ROC Curve



Definition

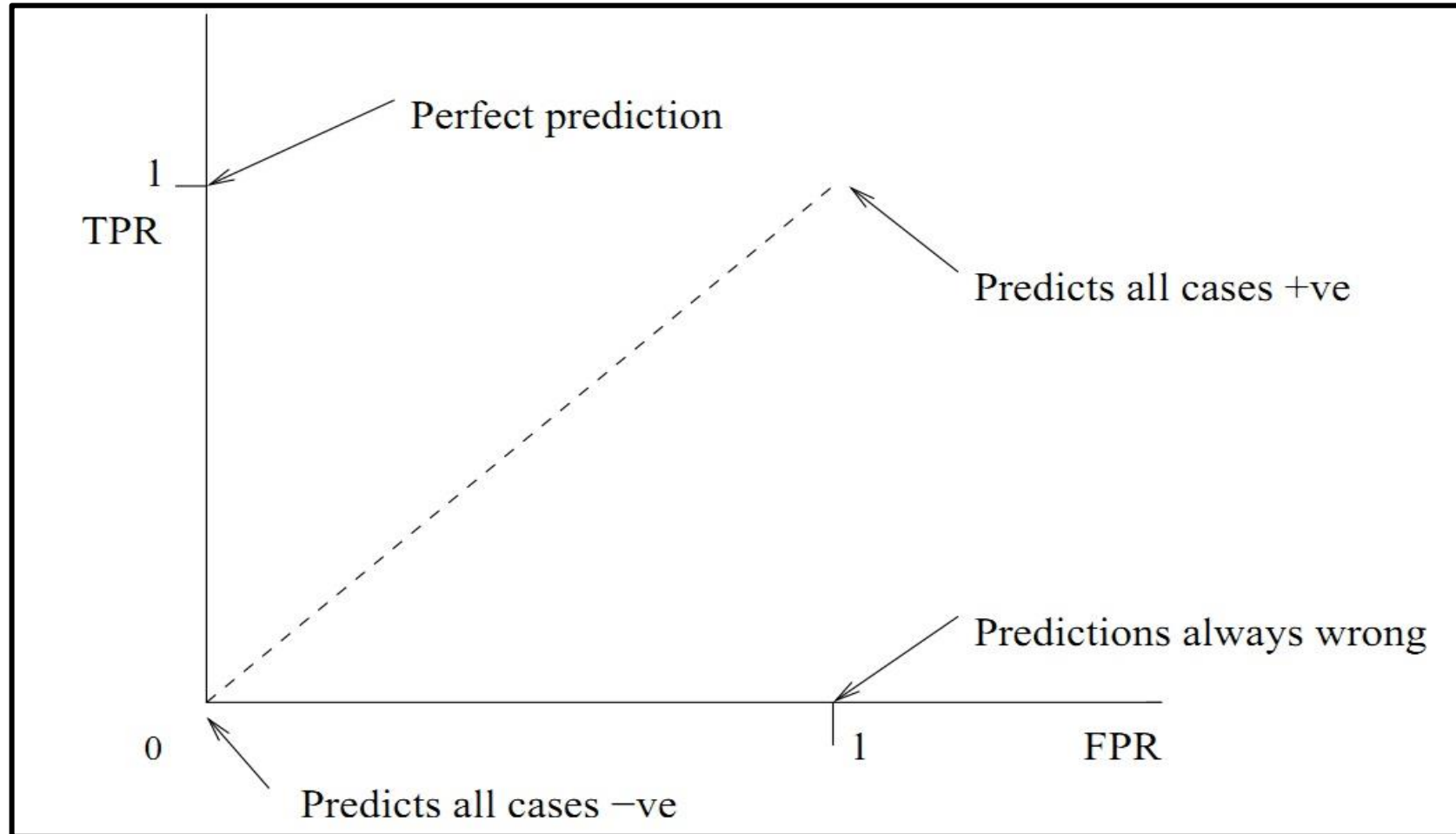
ROC Curve **summarizes** the **trade-off** between the True Positive Rate (TPR) and False Positive Rate (FPR) for a Classifier / Model using **different** Probability Thresholds



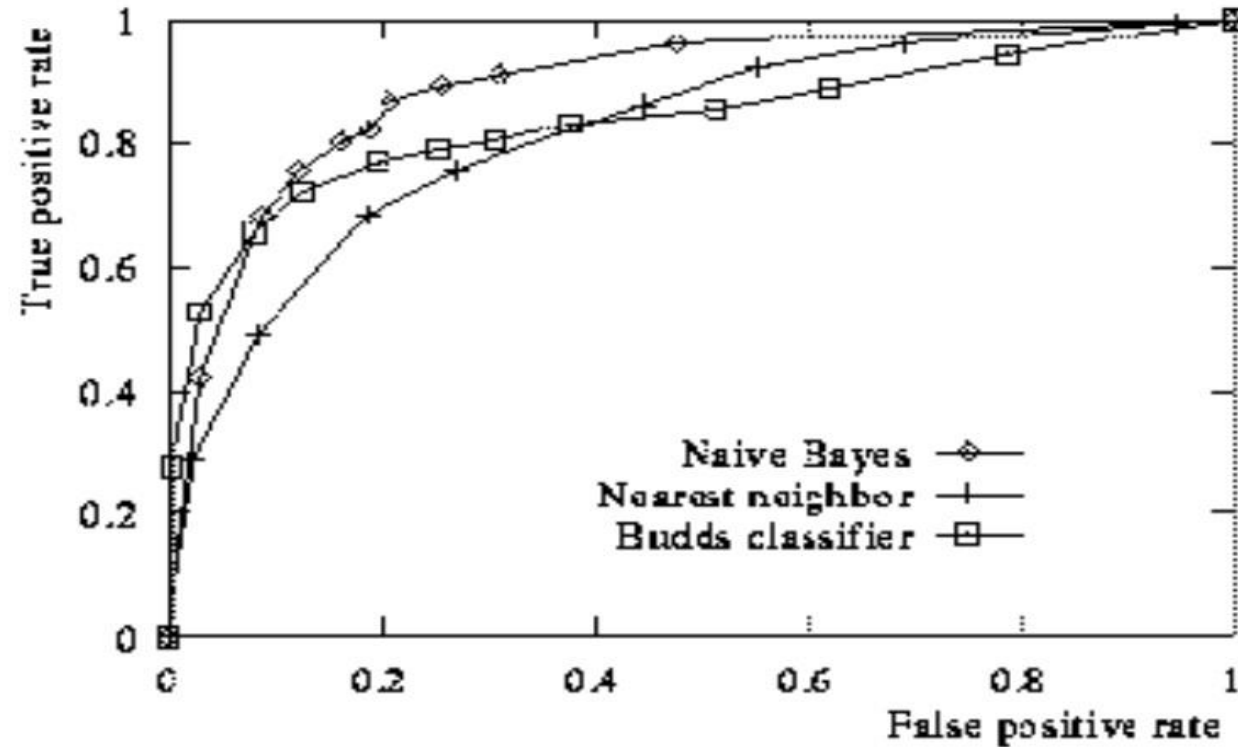
Suitable to Use

- ☐ ROC Curves are **more suitable** to use when we have
 - **Balanced Data**

ROC Curve (Cont.)



ROC Curve – Example



- ROC curves for several learning algorithms at different cost trade-offs for the task of rooftop detection in aerial images (from <http://www-csli.stanford.edu/icml2k/craft.html>)

Precision-Recall Curve

Definition

Precision-Recall Curve **summarizes** the **trade-off** between Precision and Recall (or True Positive Rate) for a Classifier / Model using **different** Probability Thresholds

Suitable to Use

- ☐ Precision-Recall Curves are **more suitable** to use when we have
 - **Highly** Unbalanced Data

Area Under the Curve (AUC)



Definition

- Area Under the ROC Curve (AUC) is defined as the **probability** that randomly chosen Positive instance is ranked above than the randomly selected Negative one



Purpose

- AUC measures the **degree of separability** between Classes
 - i.e. AUC tells how much Model is capable of **distinguishing between Classes** ?

Are Under the Curve (AUC) (Cont.)



How AUC Score is calculated?

- AUC score is computed using True Positive Rate (TPR) and False Positive Rate (FPR)**



Range of AUC Score

- Range of AUC Score is [0 - 1]**
 - 0 means All Predictions of Model are Wrong**
 - 1 means All Predictions of Model are Correct**

Are Under the Curve (AUC) (Cont.)



What is a good AUC Score?

- **AUC Score = 0.5**
 - suggests **No Discrimination** (i.e., Model **do not** have the ability to **differentiate** Positive instances from Negative ones)
- **AUC Score = 0.7 to 0.8**
 - considered Acceptable
- **AUC Score = 0.8 to 0.9**
 - considered Excellent
- **AUC Score = Greater Than 0.9**
 - considered Outstanding

ROC Graph vs AUC

ROC Graph

- ❑ ROC Graph is a **probability curve**

AUC

- ❑ AUC represents **degree of separability** between Classes